ADVANCES IN SYSTEMS IMMUNOLOGY

Course directors: Jishnu Das and Rachel Gottschalk
Monday and Wednesday 3-4:30 pm, Room B1610 Assembly Building

COURSE DESCRIPTION
This course will focus on advanced topics in Systems Immunology. Topics will include discussion of systems approaches, the general framework for applying approaches to biological problems, and specific immunologically relevant examples. The course is primarily intended to teach systems immunology techniques to graduate students in their second and third years so that they can adopt these approaches in their own research.

COURSE TOPICS

1. **Overview of systems immunology Part 1** (Singh)
   Monday 1/9/23

2. **Overview of systems immunology Part 2 (**Das/Gottschalk)
   Wednesday 1/11/23

3. **Gene regulatory networks** (Singh)
   Wedneday 1/18/23
   Singh JC Monday 1/23/23

4. **Multi-omic technologies in systems immunology** (Poholek)
   Wednesday 1/25/23
   Poholek JC Monday 1/30/23

5. **Computational approaches for cell-type deconvolution** (Chikina)
   Wednesday 2/1/23
   Chikina JC Monday 2/6/23

6. **Interpretable machine learning in human systems immunology and protein networks** (Das)
   Wednesday 2/8/23
   Das JC Monday 2/13/23

7. **Mechanistic modeling - intracellular signaling** (Faeder)
   Wednesday 2/15/23
   Faeder JC Monday 2/20/23

8. **Experimental approaches for dynamical systems** (Lee)
   Wednesday 2/22/23
   Lee JC Monday 2/27/23

Workshop 1: Designing experiments and cleaning data Wednesday 3/1/23
Workshop 2: Extracting biological meaning from gene lists Monday 3/13/23

Written proposals due: Monday 3/13/23

9. **Differentiation trajectories using computational biophysics** (Xing)
   Wednesday 3/15/23
   Xing JC Monday 3/20/23

10. **Mechanistic modeling - systems level** (Vodovotz)
    Wednesday 3/22/23
    Vodovotz JC Monday 3/27/23

11. **Modeling evolution in immunity** (Barton)
    Wednesady 3/29/23

12. **Immuno-Receptors and their engineering** (Joglekar)
    Wednesday 4/5/23
    Joglekar JC Monday 4/10/23

13. **Engineered Immune Cells** (Lohmueller)
    Wednesday 4/12/23
    Lohmueller JC Monday 4/17/23

Project oral presentations: Monday 4/24/23 and Wednesday 4/26/23

NO CLASS:
Monday 1/16/23 (Holiday, MLK observed)
Monday 3/6/23 (Holiday, Spring recess)
Wednesday 3/8/23 (Holiday, Spring recess)
Monday 4/3/23 (PMI day)
Wednesday 4/19/23 (CSHL Systems Immunology – work on projects!)

COURSE STRUCTURE
Week 1 will consist of two overview lectures. Topic-specific lectures will start on Wednesday of week 2. There will be **11** topic-specific lectures. Lectures will be 90 minutes long, including time for discussion, followed by an associated student-led JC on the next Monday (90 minutes).

ASSIGNMENTS
JC paper discussions will be led by small groups of students; students will receive their JC date assignments in week 2 of the course. *Final projects* will include a written proposal and oral presentation of the proposal, using one or more systems immunology approaches to address a research question of interest. Written proposals should be structured to include a question/hypothesis, aims, approach, and interpretations (maximum 3 pages including figures).

GRADING
Journal club participation: 50%
Final project (written and oral): 50%

Attendance is mandatory and absences must be excused in advance: email rachel.gottschalk@pitt.edu or jishnu@pitt.edu